

7.2 Singular Value Decomposition

Exam notes for $A = U\Sigma V^T$, singular values, and the four subspaces.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. The SVD writes any matrix as orthogonal input directions, nonnegative stretching, and orthogonal output directions.

For an $m \times n$ matrix,

$$A = U\Sigma V^T.$$

Here V gives orthonormal directions in $\mathbb{R}^n$, U gives orthonormal directions in $\mathbb{R}^m$, and Σ contains the singular values.

2 What the SVD Says

$$Av_i = \sigma_i u_i.$$

This means: the right singular vector v_i is sent to the left singular vector u_i , scaled by $\sigma_i \geq 0$.

Object	Lives in	Meaning
v_i	$\mathbb{R}^n$	Orthonormal input direction.
u_i	$\mathbb{R}^m$	Orthonormal output direction.
σ_i	Nonnegative number	Stretch factor from v_i to u_i .
Σ	$m \times n$ diagonal shape	Stores singular values in decreasing order.

3 How to Compute It

The right singular vectors come from the symmetric matrix $A^T A$:

$$A^T A v_i = \sigma_i^2 v_i.$$

So the singular values are

$$\sigma_i = \sqrt{\lambda_i(A^T A)}.$$

For each $\sigma_i > 0$, compute the matching left singular vector by

$$u_i = \frac{A v_i}{\sigma_i}.$$

If U or V is not complete, add orthonormal vectors to finish the basis.

4 Geometry

The SVD is the clean geometric order:

$$x \xrightarrow{V^T} \text{rotate/reflection} \xrightarrow{\Sigma} \text{stretch} \xrightarrow{U} \text{rotate/reflection}.$$

The singular values show the principal stretches of the matrix.

5 Rank and the Four Subspaces

If $\sigma_1, \dots, \sigma_r > 0$ and the rest are zero, then $r = \text{rank}(A)$.

Subspace	Basis from the SVD
Row space $C(A^T)$	$v_1, \dots, v_r$.
Nullspace $N(A)$	Remaining v_i with $\sigma_i = 0$.
Column space $C(A)$	$u_1, \dots, u_r$.
Left nullspace $N(A^T)$	Remaining u_i .

6 SVD vs Eigenvalue Decomposition

Feature	Eigenvalue decomposition	SVD
Basic form	$A = S\Lambda S^{-1}$	$A = U\Sigma V^T$
Works for	Diagonalizable square matrices	Every matrix
Numbers	Eigenvalues may be negative or complex	Singular values are nonnegative real numbers
Directions	Same space in and out	Input directions v_i , output directions u_i
Best use	Dynamics and powers	Geometry, rank, least squares, approximation

7 Symmetric Case

For a symmetric positive definite matrix,

$$A = Q\Lambda Q^T$$

already has orthonormal eigenvectors and positive eigenvalues. In this special case, the SVD agrees with the eigenvalue decomposition.

For a symmetric matrix with negative eigenvalues, singular values use absolute sizes, not signs.

8 Common Mistakes

Mistake	Correct exam habit
Treating singular values as eigenvalues of A .	Singular values come from $A^T A$.
Allowing negative singular values.	Always take $\sigma_i \geq 0$.
Using $u_i = Av_i/\sigma_i$ when $\sigma_i = 0$.	Only divide by positive singular values.
Forgetting rectangular dimensions.	Σ has the same shape as A .
Mixing up U and V .	V belongs to the input side; U belongs to the output side.

9 Final Exam Checklist

Checklist.

1. Compute $A^T A$.
2. Find eigenvalues and orthonormal eigenvectors of $A^T A$.
3. Set $\sigma_i = \sqrt{\lambda_i}$.
4. Put the v_i 's into V .
5. Compute $u_i = Av_i/\sigma_i$ for positive σ_i .
6. Complete orthonormal bases when needed.
7. Write $A = U\Sigma V^T$ with matching dimensions.
8. Use zero singular values to identify nullspaces.